

GrounUp Enterprise Specification

Phase 99 - GrounUp Enterprise Master Specification

Version 1.0.0 | Authoritative Umbrella Specification | Product Certification: NOT CERTIFIED

1. Purpose and Authority

Phase 99 is the umbrella specification for the complete GrounUp Enterprise Specification. It incorporates the verified Phase 01-32 governed baseline by reference while preserving the detailed source language, IDs, tests, controls, dependencies, and evidence obligations of each phase.

2. Incorporated Governed Baseline

Phase 99 incorporates 32 governed phases, 8,600 unique requirement IDs, 8,858 unique test IDs, 125 directed dependency edges, and 10 dependency-aware implementation waves. The verified GES v1.0 Master Repository snapshot is included with this package.

3. Precedence and Conflict Rule

A Phase 99 summary, registry, workbook, diagram, AI instruction, or developer guide is a derived control artifact and may not silently replace the detailed governing source language. Apparent conflicts are stop conditions for the affected implementation path until a controlled change or architecture decision resolves them.

4. Source-of-Truth Boundaries

Phase 24 supplies the reusable core platform foundation. Phase 25 governs master-library versions and snapshots. Phase 26 is authoritative for deterministic estimating calculations. Phase 27 governs AI construction intelligence. Phase 28 security applies across every later layer. Phase 29 analytics is analytical, not a hidden transaction authority. Phase 30 entitlement never overrides security authorization. Phase 31 governs production release/recovery. Phase 32 alone governs real acceptance and production certification.

5. Implementation Governance

Engineering follows dependency-aware implementation waves, not a naive assumption that every numbered phase is a coding step. Security controls are implemented with the foundation rather than postponed. Every feature must trace requirement IDs to code/configuration, tests, migrations where applicable, security/audit controls, observability, and release evidence.

6. AI and Developer Execution

AI coding agents and human developers must read the exact governing artifacts before implementation. They may not invent missing requirements, weaken tests or security controls, silently alter locked source packages, or claim expected future evidence as completed. Completion reporting must include requirements, changed files, migrations/configuration, test results, security/tenant impact, API/event impact, observability impact, evidence, blockers, and gate PASS/FAIL.

7. Deployment and Release

The governed promotion path is Local -> Development -> Integration/Test -> Staging/UAT -> Pilot -> Production Candidate -> Production -> Hypercare -> Steady State, or an approved equivalent controlled path. Production promotion uses immutable tested artifacts and exact configuration/migration references. Blocking security, tenant, data, deterministic calculation, migration, recovery, or acceptance failures stop promotion.

8. Architecture Decisions

Provider-neutral boundaries remain neutral until ADRs approve cloud/region, runtime, database/tenancy, cache/queue/object/search, identity/SSO/SCIM, secrets/KMS/certificates, CI/CD/artifacts, observability/SIEM/on-call, analytics, AI providers/routing, commercial providers, and backup/DR decisions.

9. Validation and Certification

Phase 99 package completion certifies only that the master specification package has passed its own artifact QA. It does not prove real product UAT, pilot deployment, migration/cutover, production verification, disaster recovery, hypercare, or customer acceptance. Product production certification requires Phase 32 executed evidence and authorized signoff for the exact release scope.

10. Change and Version Control

Original incorporated source ZIPs are immutable. Corrections or semantic changes create versioned governed changes with impact analysis, dependency updates, and revised manifests. Existing IDs remain stable unless an explicitly governed versioning rule says otherwise.

Master-Control Requirements - Critical Rules

P99-REQ-001 - Master Authority: Phase 99 is the umbrella specification and incorporates the governed Phase 01-32 source packages by reference; it does not supersede their detailed requirements unless an explicitly versioned approved change says so.

P99-REQ-002 - Immutable Sources: The Phase 01-32 governed ZIPs and the GES v1.0 Master Repository snapshot are immutable source artifacts for this Phase 99 version.

P99-REQ-003 - No Silent Reconciliation: When two governed sources appear inconsistent, implementation must stop on the affected path and open a controlled architecture/change decision; an AI or developer may not silently choose a preferred interpretation.

P99-REQ-004 - Requirement Incorporation: All 8,600 unique governed requirement IDs in the verified master registry are incorporated by reference into Phase 99.

P99-REQ-005 - Test Incorporation: All 8,858 unique governed test IDs in the verified master registry are incorporated by reference into Phase 99.

P99-REQ-006 - Phase Identity: Existing Phase 01-32 IDs retain their original identity and home phase; Phase 99 does not renumber them.

P99-REQ-007 - Traceability Minimum: Every implemented feature must trace governing requirement IDs to code/configuration, tests, migrations where applicable, security controls, observability and release evidence.

P99-REQ-008 - Change Control: A semantic change to a governed requirement requires explicit version/change evidence and impact analysis; editing a derived master summary is not sufficient.

P99-REQ-009 - Phase 22 Lock: Phase 22 remains the final reconciliation/enterprise-lock package for the governed specification baseline before software implementation foundations.

P99-REQ-010 - Phase 24 Foundation: Phase 24 core identity, tenancy, data, API, audit and workflow contracts are the reusable implementation foundation for platform modules.

P99-REQ-011 - Phase 25 Authority: Published Phase 25 master-library/service/resource/material/equipment/assembly/rate versions and snapshots are governed data authorities.

P99-REQ-012 - Phase 26 Authority: Phase 26 deterministic estimating/cost/production/crew/equipment/haul/markup calculations are authoritative for estimate math.

P99-REQ-013 - AI Math Boundary: AI may explain, prepare inputs for, or review deterministic calculations but may not silently replace Phase 26 authoritative calculation results.

P99-REQ-014 - Phase 27 AI Boundary: Phase 27 AI/copilot/agent behavior must remain evidence-grounded, permission-scoped, bounded by tool policy, and human-gated for protected actions.

P99-REQ-015 - Phase 28 Security: Phase 28 security, tenant isolation, least privilege, zero-trust, secrets, audit and secure-SDLC controls apply across every implemented module and provider integration.

P99-REQ-016 - Security vs Entitlement: Phase 30 commercial entitlement may restrict product capability but may never expand access beyond Phase 28 authorization.

P99-REQ-017 - Phase 29 Authority: Phase 29 analytics may derive certified analytical outputs but may not become the hidden transaction authority for operational systems.

P99-REQ-018 - Phase 31 Release Rule: Production promotion uses immutable tested artifacts with exact configuration/migration references and release evidence.

P99-REQ-019 - Phase 32 Truth Rule: No artifact, code completion, demo, test plan, or expected future result may be represented as actual UAT, pilot, migration, production verification, or production certification.

P99-REQ-020 - Production Certification: Production certification requires executed Phase 32 evidence and authorized signoffs for the exact release/artifact/configuration scope.

P99-REQ-021 - Tenant Isolation: Tenant context must be explicit and enforced across APIs, repositories, cache, files, search, events, analytics, AI retrieval/tools, exports and commercial records.

P99-REQ-022 - Default Deny: Protected actions are denied unless explicit governing identity/authorization/security policy permits them.

P99-REQ-023 - Secrets: Live secrets, credentials and private keys are not stored in source packages, prompts, ordinary configuration, logs or runbooks.

P99-REQ-024 - Schema Change: Schema changes are versioned, migration-tested and assessed for backward compatibility, tenant safety, analytics and rollback/forward-recovery impact.

P99-REQ-025 - API Compatibility: Material API/event contract changes require versioning/compatibility review and downstream impact analysis.

P99-REQ-026 - Provider Neutrality: Provider-neutral specifications remain neutral until a governed ADR approves a provider binding with rationale, security review and exit strategy.

P99-REQ-027 - Architecture Decisions: Cloud, runtime, data, identity, secrets, CI/CD, observability, analytics, AI provider, commercial provider and DR choices require ADRs before production binding.

P99-REQ-028 - Build Order: Engineering implementation follows dependency-aware waves rather than simply treating all phase numbers as a linear coding order.

P99-REQ-029 - Security From Start: Phase 28 controls are implemented with the platform foundation, not postponed until after business features are complete.

P99-REQ-030 - Golden Tests: Authoritative formulas, entitlement combinations, certified metrics and migration reconciliations use controlled golden tests where the governing phases require them.

P99-REQ-031 - Evidence Before Pass: A gate cannot be marked passed without required evidence; unresolved blocking failures require resolution or an explicitly authorized exception/risk acceptance.

P99-REQ-032 - No Test Weakening: Tests and controls may not be weakened, skipped or relabeled simply to make an implementation appear complete.

P99-REQ-033 - AI Coding Agent Contract: AI coding agents must report requirements implemented, files changed, migrations/config, tests/results, security/tenant impact, API/event impact, observability impact, evidence and gate PASS/FAIL.

P99-REQ-034 - AI Source Discipline: AI coding agents must read the exact governing source artifacts before coding and may not implement solely from a summary or memory.

P99-REQ-035 - Deployment Promotion: Promotion path is Local -> Development -> Integration/Test -> Staging/UAT -> Pilot -> Production Candidate -> Production -> Hypercare -> Steady State unless an approved architecture/release decision defines an equivalent controlled path.

P99-REQ-036 - Stop Conditions: Promotion stops for unaccepted blocking security, tenant-isolation, data-integrity, deterministic-calculation, migration, recovery or acceptance failures.

P99-REQ-037 - Backup/Restore: Backup success alone is not proof of recoverability; critical services require restore validation according to Phase 31.

P99-REQ-038 - RTO/RPO: Critical recovery objectives must be demonstrated by exercises or carry an authorized exception.

P99-REQ-039 - Operational Ownership: Critical production services have named owners, runbooks, observability, escalation and recovery responsibilities.

P99-REQ-040 - White Label Boundary: White-label configuration changes branding and permitted commercial presentation, not platform security identity or tenant isolation.

P99-REQ-041 - Master Manifest: Every released Phase 99 package includes a machine-readable SHA-256 manifest and source provenance.

P99-REQ-042 - Release Archive: The release archive preserves exact governed source packages, master registries, implementation controls, known issues, risk decisions and evidence references.

P99-REQ-043 - Derived Artifact Rule: Derived registries, diagrams, workbooks and summaries improve navigation but do not override the incorporated source package language.

P99-REQ-044 - Exception Expiry: Security, commercial, quality and release exceptions are explicit, owned, evidenced and time/review bounded where the governing control requires it.

P99-REQ-045 - No False Completion: Phase 99 is complete as a master specification when its package/QA gates pass; this does not mean the software product itself is production-certified.

Implementation Waves

Wave	Name	Phases	Focus	Exit Gate
0	Governed baseline reconciliation	01-22	Load governed specifications, final reconciliation and enterprise	Baseline implemented, SDR/Playbook backdoored, no out for
1	Repository & architecture foundation	23	Repository bootstrap, architecture boundaries, contracts, traceability	Repository level target A/B baseline acceptance.
2	Core platform + security foundation	24 + 28 baseline	Tenant context, identity, authorization, audit, workflow, configuration	Tenant resources and foundation security foundation pass
3	Master data and deterministic engineering	25-26	Service catalog, resources, assemblies, rates, snapshots, production	Core cost/benefit and single shot reproducibility pass
4	Operational domain applications	07-19 guided by 02-06	Project, CRM, workforce, finance, procurement, fleet, HSE, documents, GIS/GPS, selection	Domain, GIS/GPS, selection commercial workflow
5	AI construction intelligence	06 + 27	Evidence-grounded copilot/agents over governed retrieval, Phase 25 ending and Phase 26 security	AI construction intelligence pass
6	Enterprise analytics	13 + 29	Warehouse/lakehouse, quality, lineage, semantic metrics, reporting	Optimized point in time information leakage/security pass
7	SaaS commercialization & white label	30	Plans, subscriptions, usage, billing, entitlements, onboarding, support	Commercialization and white label gates pass
8	Production infrastructure & SRE	20 + 28 + 31	Approved provider binding, IaC, CI/CD, observability, backup/recovery	Release to Production/DR security gates pass.
9	UAT, pilot, migration & certification	32	Execute real acceptance, pilot, migration/cutover, production verification	Execution acceptance and signoff, executed evidence

Incorporated Phase Registry

Phase	Title	Files	Integrity
P01	Governance	51	PASS
P02	Core Platform	55	PASS
P03	Enterprise Data Model	53	PASS
P04	Master Library Registry	39	PASS
P05	Estimating Cost Intelligence	55	PASS
P06	AI Construction Intelligence	40	PASS
P07	Enterprise Project Management	44	PASS
P08	Enterprise CRM Sales Customer Experience	44	PASS
P09	HR Workforce Talent Management	46	PASS
P10	Enterprise Financial Management Accounting	47	PASS
P11	Enterprise Procurement Supply Chain Inventory	44	PASS
P12	Enterprise Fleet Asset Maintenance Telematics	52	PASS
P13	Enterprise Business Intelligence Decision Intelligence	37	PASS
P14	Enterprise HSE Safety Environmental Quality Compliance	37	PASS
P15	Enterprise Document Drawing Specification Information Management	37	PASS
P16	GIS Survey Reality Capture Drone Operations	37	PASS
P17	Machine Control GPS Field Automation	37	PASS
P18	Enterprise Scheduling Resource Planning	37	PASS
P19	Contract Change Order Claims Commercial Management	37	PASS
P20	Enterprise Integration Deployment Testing Governance Production Readiness	37	PASS
P21	Master Enterprise Consolidation Traceability Implementation Baseline	33	PASS
P22	Final Reconciliation Enterprise Lock Developer AI Execution Package	28	PASS
P23	Software Development Foundation Reference Architecture Repository Bootstrap	41	PASS
P24	Core Platform Implementation Identity Tenancy Data API Audit Workflow	42	PASS
P25	Master Library Service Catalog Resource Material Equipment Assembly Rate Management	46	PASS
P26	Estimating Cost Production Crew Equipment Haul Markup Execution Engine	46	PASS
P27	AI Construction Copilot Autonomous Agents Plan Spec Intelligence Human Approval	46	PASS
P28	Enterprise Security Identity Zero Trust Privacy Audit Threat Detection	47	PASS
P29	Enterprise Reporting Data Warehouse Lakehouse Semantic Analytics AI Ready Data	44	PASS
P30	SaaS Commercialization Billing Licensing Entitlements White Label Customer Operations	33	PASS
P31	Production Deployment Cloud Infrastructure DevOps CICD Observability DR SaaS Operations	42	PASS
P32	Enterprise Validation UAT Pilot Migration Launch GoLive Certification	31	PASS

Phase 99 Quality Gates

P99-QG-01 - Master Source Integrity: Master repository ZIP and source phase provenance verify (Required)

P99-QG-02 - Phase Completeness: Exactly Phase 01-32 incorporated (Required)

P99-QG-03 - Requirement Registry: 8,600 unique source requirement IDs incorporated (Required)

P99-QG-04 - Test Registry: 8,858 unique source test IDs incorporated (Required)

P99-QG-05 - Definition Conflicts: 0 canonical requirement/test definition conflicts (Required)

P99-QG-06 - Dependency Registry: Dependency map parses and references valid phases (Required)

P99-QG-07 - Build Sequence: All 32 phases represented in build sequence (Required)

P99-QG-08 - Implementation Waves: Dependency-aware implementation waves documented (Required)

P99-QG-09 - AI/Developer Protocol: Coding-agent and developer controls included (Required)

P99-QG-10 - Deployment Roadmap: Environment promotion and stop conditions documented (Required)

P99-QG-11 - Security Boundary: Phase 28 inheritance explicitly preserved (Required)

P99-QG-12 - Deterministic Authority: Phase 26 calculation authority explicitly preserved (Required)

P99-QG-13 - Certification Boundary: Phase 32 real-execution boundary explicitly preserved (Required)

P99-QG-14 - CSV/JSON Parse: All Phase 99 control CSV/JSON files parse (Required)

P99-QG-15 - Workbook QA: Spreadsheet formula scan and rendered dashboard pass (Required)

P99-QG-16 - DOCX QA: DOCX renders cleanly (Required)

P99-QG-17 - PDF QA: PDF renders cleanly (Required)

P99-QG-18 - Manifest QA: Phase 99 SHA-256 manifest verifies (Required)

P99-QG-19 - ZIP Integrity: Final Phase 99 ZIP CRC test passes (Required)

P99-QG-20 - Product Certification Truth: Package remains NOT PRODUCTION CERTIFIED absent Phase 32 evidence (Required)